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MICROESTRUCTURA Y SISTEMAS DE TRADING



PROYECT 003 ADVANCED TRADING STRATEGIES: PAIRS TRADING

Presents:

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1. Introduction

Trading consists of buying and selling financial instruments, and the goal of this is making profits from price changes. In general, this activity requires a deep market analysis, identifying trends and proper risk management. Therefore, it is very important establishing rules that reduce the influence of subjectivity on decision-making.

This project presents the design, implementation and evaluation of a statistical arbitrage strategy based on pairs trading, where is needed to find cointegrated assets, after calculating the spread, get mean reversion dynamics. The pair used in this project is Microsoft (MSFT) and Intuit (INTU) and it was selected from the technology sector from the U.S. market based on their long-term high correlation and successful results statistic cointegration tests.

This strategy uses a Kalman Filter to estimate dynamic hedge ratios in real time following the Powell's Sequential Decision Analysis (SDA). In addition, a backtesting was performed using 15 years of historical daily data and it was divided into 60% training, 20% testing and 20% validations sets. In addition, the strategy will start with an initial capital of \$1,000,000 USD, and in each trade 80% of the current capital will be invested, dividing 40% on each stock.

2. Strategy Description and Rationale

2.1 Overview of a pairs trading approach

First, defining a linear regression where the objective is to predict a stock price Y using another stock price X : $Y = \alpha + \beta X + \epsilon$, the spread of the pair is defined as the residuals ϵ from the regression. With this, it is needed to run some tests to validate that the pair in question is cointegrated, and once it is, it is possible to get open and close position signals for the trading strategy.

2.2 Why cointegration indicates arbitrage opportunity

Cointegration is the statistical principal that validates the long-term profitability of this strategy. It implies that the pair of stocks are two non-stationary time series, and if the linear combination between them (the spread) is in fact stationary, then the pair is statistically cointegrated. With this, when the spread is deviated from the mean, it is expected to revert over time. In financial markets statistically cointegration isn't enough; it is also needed an economic relationship, such as shared market exposure to a sector, similar fundamentals, or operating in same sector or serving similar markets.

Pairs trading is a statistical arbitrage strategy that take advantage on temporary deviations on the spread between to cointegrated assets. The main idea is having the standardized spread of the pair, which follows a normal standard distribution $\mathcal{N}(0, 1)$, and a threshold number of standard deviations. Defining the threshold at ± 1.5 standard deviations, if the standardized spread exceeds 1.5 standard deviations, a shorting signal for the spread is issued and this would imply shorting Y and go long on X . On the other hand, if the standardized spread gets below -1.5 standard deviations, a long signal for the spread is issued and this would imply go long on Y and shorting X .

The rationale of this strategy is that the standardized spread statistically must be a stationary series. Taking this into account, when the spread exceeds the threshold, it is expected a mean reversion process. This process generates profits independent of the general direction of the market and this property makes it highly attractive for algorithmic implementation and portfolio-level risk control.

2.3 Justification for Kalman filter use in dynamic hedging

Traditional pairs trading models often rely only on a static Ordinary Least Squares (OLS) regression to estimate α and important the hedge ratio β between to assets. However, in the stock market nothing is static, and prices are changing constantly so the Kalman Filter addresses this limitation by providing a recursive Bayesian estimation that updates the hedge ratio dynamically in real time. This recursive approach ensures the model doesn't lose sensitivity to new information or overreacting to noise on the market.

2.4 Expected market conditions for strategy success

The expected market conditions for this strategy success include the following points:

- Stable macroeconomic conditions so it favors predictability and reduce big shocks on the market.
- Moderate volatility, since little volatility may reduce trading opportunities.
- High liquidity ensures the closing positions on a desirable price.
- Low transaction, costs since depending on the pair or the strategy there could be la lot of trades in short periods of time.

3. Pair selection methodology

3.1 Correlation screening criteria and results

The first step in a pair selection process is to identify stocks that share a historical correlation. To measure this, a 50-day rolling correlation was used. This involves creating a 50-day window over the data, calculating the correlation of the daily returns of both stocks, and finally, traversing the window day by day and calculating the correlation each time the window is moved. The results are saved, and at the end, the average of all calculations is computed.

A threshold of $\rho > 0.5$ was set to take out pairs with weak correlations, since a large database of assets separated by sector was used to find out the best pair on the biggest companies in the U.S. market. This threshold doesn't really ensure cointegration, nevertheless it is necessary as a first condition for finding it and it is helpful as a first filter before other tests.

Between all the combinations by sector, the pairs that survived this test were the following ones:

PAIR	SECTOR	MEAN ROLLING CORRELATION
MSFT-INTU	Information Technology	0.510284
ORCL-ACN	Information Technology	0.512785
IBM-ACN	Information Technology	0.5372
DHR-SYK	Health Care	0.520726
SYK-TMO	Health Care	0.542778
MS-COF	Financials	0.62865
BK-SCHW	Financials	0.710108
BLK-SCHW	Financials	0.656435
BLK-COF	Financials	0.607868
USB-COF	Financials	0.671076
SCHW-COF	Financials	0.605467
MET-COF	Financials	0.615327

3.2 Cointegration testing (Engle-Granger method)

3.2.1 Augmented Dickey-Fuller (ADF) test.

The ADF test is a statistical procedure to determine whether a time series is stationary or contains a unit root. As every statistical test, there must be two hypotheses formulated to validate the results. Formally, the ADF test evaluates the following hypotheses:

- H_0 (Null hypothesis): The series is non-stationary.
- H_1 (Alternative hypothesis): The series is stationary.

For deciding, the p-value (probability that measures the strength of the evidence against the null hypothesis) and a significance level α is needed. If $p < \alpha$ the null hypothesis must be rejected and conclude that the series is in fact stationary. If $p > \alpha$ there is no evidence to reject the null hypothesis, therefore the series is stationary.

3.2.2 Engle-Granger Method

First an ADF test is performed on both close price series of the stocks. And the results must show that the series aren't stationary. Then by running an OLS regression between the asset prices is performed and formulated as said on the beginning of this report: $Y = \alpha + \beta X + \epsilon$, where Y and X represent the daily closing prices of two assets. Then an ADF test is performed on the residuals ϵ to determine whether the spread is stationary or not. The results must show that the regression residuals are in fact stationary. Once both conditions are validated, it is possible to conclude that the pair is in fact cointegrated.

3.3 Statistical evidence for selected pairs

Once the Engle-Granger Method was used to find cointegrated pairs, the obtained results are the following:

PAIR	SECTOR	ADF PVALUE	CORR.	COINTEGRATED
MSFT-INTU	Information Technology	0.000077	0.510284	Yes
ORCL-ACN	Information Technology	0.00549	0.512785	Yes
IBM-ACN	Information Technology	0.00628	0.5372	Yes
DHR-SYK	Health Care	0.00577	0.520726	Yes
SYK-TMO	Health Care	0.00216	0.542778	Yes
MS-COF	Financials	0.00703	0.62865	Yes
BK-SCHW	Financials	0.00069	0.710108	Yes
BLK-SCHW	Financials	0.00762	0.656435	Yes
BLK-COF	Financials	0.00375	0.607868	Yes
USB-COF	Financials	0.00373	0.671076	Yes
SCHW-COF	Financials	0.00769	0.605467	Yes
MET-COF	Financials	0.00534	0.615327	Yes

With this results the pair MSFT-INTU is selected as the best cointegrated pair since the p-value obtained on the ADF test is the smallest one of all pairs studied. However, these tests don't ensure economical relationship at all, so a much deeper analysis must be done to validate that makes sense and it's rational to use it.

3.4 Economical analysis between Microsoft and Intuit

Microsoft (MSFT) and Intuit (INTU) are two companies with strong economic and fundamental relationship since both compete on the software and technology sector, particularly on the business productivity, cloud computing and financial software solutions. Both companies have small and medium sized companies as clients that depend on digital tools for accounting, operations and business management,

On one hand, MSFT dominates on the productivity area with software like Office 365 and cloud infrastructure with Azure. However, INTU leads the financial management solutions. Both are exposed to similar macroeconomic cycles, technology risks and trends, and adoption of cloud-based systems. This leads to high correlation and potential long-term cointegration on their stock prices.

3.5 Standardized prices of the best pair and spread over time

After selecting the best pair for this strategy, the standardized prices of both stocks are shown on illustration 1, and it can be seen that there is an important cointegration and correlation. Also based on how the strategy work, every time the standardize prices cross one over the other there should be a trade opportunity, and on this pair, there are several crosses over time, which means that there are great chances to get good results.

On the other hand, on illustration 2, it is showed the spread over time on train set, and on the test and validation test is the predicted spread using the Kalman Filter. It's evident that once the predictions started the volatility on the spread increases, but that's the reason why the trading signals are issued from the standardized spread and using a rolling window for calculating the μ and σ for the standardization process, which is detailed later in the report.

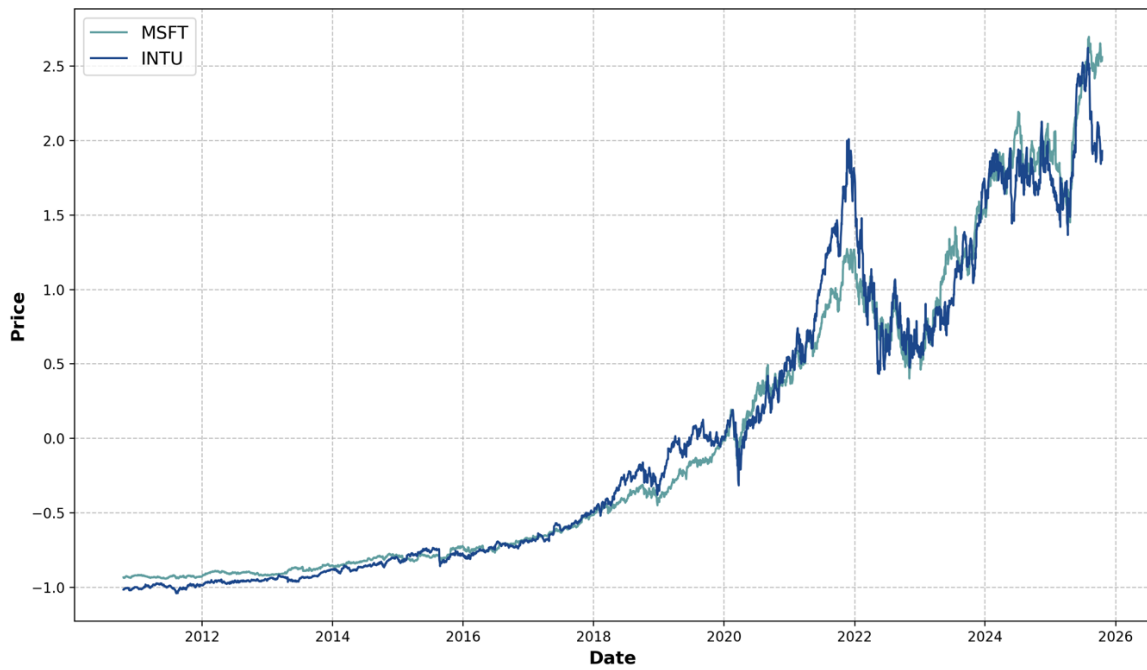


Illustration 1: Plot of the standardized MSFT and INTU price series. (Source: own elaboration).

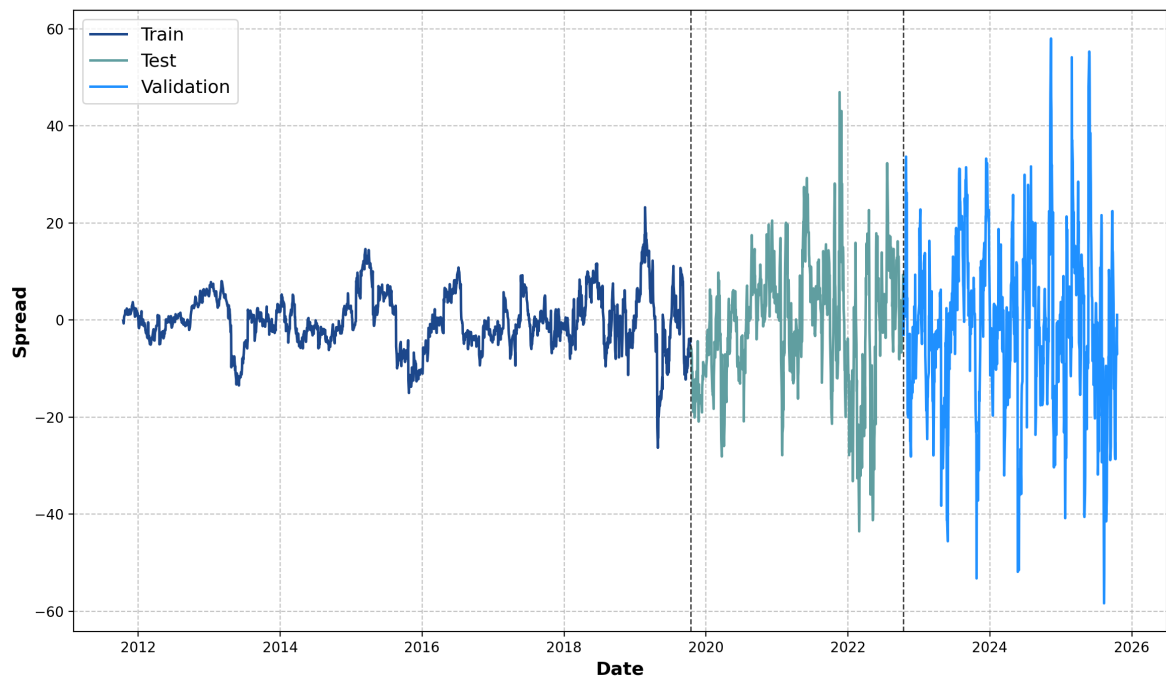


Illustration 2: Plot of the spread between MSFT and INTU separated on train, test and validation (last two calculated with the Kalman Filter model). (Source: own elaboration).

4. Sequential Decision Analysis (SDA) Framework

4.1 What is SDA and why is important on this project

The Sequential Decision Analysis (SDA) is a framework for problems for modeling decisions that must be made repeatedly over time, under uncertainty and with new information arriving in a sequential way. The use in this project is for treating this trading strategy as a repeated learning process, where each trading point on time represents a new decision time point, instead of treating it as a static optimization.

On SDA, each period implies observing new information, update believes, choose an action, experience some noise and move to a new state. This can be transferred to the pairs trading strategy with the Kalman Filter implementation. Price would arrive daily, then update believes in the hedge ration using the Kalman Filter, make a trading decision and face uncertainty with the spread not being able to reverse.

4.2 Detailed mathematical formulation of state-space model

4.2.1 State S_t

The state S_t of the system at time t contains all the necessary and sufficient information to model our system from time t onward. S_0 is the initial state that captures deterministic parameters that never change, initial values that do change and beliefs about quantities or parameters that aren't known perfectly (like probability distributions).

4.2.2 Decision variables

Defining x_t as a decision at time t , in this problem there are 3 categorical decisions: open a long position on the spread (1), hold position (0) or short spread (-1) the spread. The decisions are made based on a policy, which is denoted as $X^\Pi(S_t)$. Also, a decision $x_t = X^\Pi(S_t)$ is feasible at time t and the policy would compute the decision for the signal of the day using additionally some tunable parameters $\theta \in \Theta$.

4.2.3 Exogenous Information

Any new information that is will be known on time $t + 1$ will be denoted as W_{t+1} and is the exogenous information on the model, since the source is from the outside of our system. For example, in this strategy the next day prices X_{t+1} and Y_{t+1} and the market noise may be the exogenous information.

4.2.4 Transition function

The transition function is denoted as $S_{t+1} = S^M(S_t, x_t, W_{t+1})$. This function must be able to capture changes, update to new information and update the beliefs about the uncertainty. In this project the transition is composed by the prediction and updating process on the Kalman Filter detailed further and implemented on the backtesting of the strategy

4.2.5 Objective function

Defining the contribution on time t as $C(S_t, x_t, W_{t+1})$, the objective would be to maximize the cumulative contribution over a time horizon T . Also, the objective function must depend on the initial state S_0 . This can be expressed as:

$$\max_{\Pi} \mathbb{E} \left[\sum_{t=0}^{T-1} C(S_t, x_t^{\Pi}, W_{t+1}) \right]$$

Remembering that Π represents the policy, in this case that will be the Z-score thresholds that make the daily signals, that also are the decision on each time t . In this project the policy can be evaluated with risk-adjusted metrics such as Sharpe, Sortino or Calmar ratios, and at the end get some trade statistics like win rate or the profit factor.

4.3 The six-step modeling process

4.3.1 The narrative

The problem being modeled is to dynamically estimate the hedge ratio between two cointegrated assets, in this case MSFT and INTU for a pairs trading strategy. On every trading day the strategy must be able to decide to open close or hold a spread position based on the observed data. Unlike using a static regression model, this system must adapt to new information. The Kalman Filter is the key element, since will provide a sequential decision-making framework allowing the model to learn and update its beliefs by itself as new information arrives.

The first thing to define will be the beliefs. From the Kalman Filter $(\hat{\alpha}_t, \hat{\beta}_t, P_t)$ where $\hat{\alpha}_t$ and $\hat{\beta}_t$ are the estimations of the linear regression and P_t is the estimation error covariance matrix, and quantifies the model's uncertainty about the current state $(\hat{\alpha}_t, \hat{\beta}_t)$. A higher P_t will indicate less confidence and a lower P_t a better stability and smoother estimates. Other beliefs for the model are (μ_t, σ_t) which are the mean and standard deviation of the spread on time t which are calculated with a rolling window of 252 days (1 stock market year).

Next, the portfolio state must include the following:

- Current cash at time t .
- Open positions at time t with their respective number of shares, entry prices, commission and borrow costs.
- Current prices of each stock X_t and Y_t at time t .

4.3.2 Core elements of the problem

First, the metric that is primarily being tried to impact is the Sortino ratio, since this is a risk-adjusted metric that helps to maximize expected returns over the downside risks of a portfolio. This metric will be detailed further. Also as said before, the decisions that will be made are open a long position on the spread (1), hold position (0) or short spread (-1) between MSFT and INTU, and they are based on the policy.

The main sources of uncertainty include the evolution of the stock prices, that the parameters won't be longer the best for the estimation of $\hat{\alpha}_t$ and $\hat{\beta}_t$ and the market noise that may affect short term reversion processes. Also, before starting, there are uncertainties, such as if the pair will have a truly long term cointegration. And finally, the exogenous information that will arrive may be the unknown stock prices or future market noise or events that could affect the strategy.

4.3.3 The mathematical model

The Kalman Filter will provide the recursive estimation structure on this problem. Although this will be detailed further on this report, is necessary to know that the basic structure is as follows:

$$w_t = Fw_{t-1} + v, \quad v \sim \mathcal{N}(0, Q)$$

$$y_t = X_t w_t + \epsilon, \quad \epsilon \sim \mathcal{N}(0, R)$$

In this structure $w_t = [\alpha_t, \beta_t]^T$ represents the state vector containing the model coefficients. Then F is the transition matrix (identity in this case) and Q and R are the process and observation noise covariance matrices respectively. The objective in this optimization will be finding a trading policy $X^\Pi(S_t)$ that maximizes:

$$\max_{\Pi} \mathbb{E} \left[\sum_{t=0}^{T-1} C(S_t, x_t^\Pi, W_{t+1}) \right]$$

4.3.4 Uncertainty model

1. The market noise: unpredictable movements in MSFT and INTU prices that will affect the spread over time.
2. The optimal parameter changing with the Kalman Filter these uncertainties are modeled by changing dynamically $(\hat{\alpha}_t, \hat{\beta}_t)$ through covariance matrices of the process noise Q and measurement noise R that are tuned by hand to balance the stability.

4.3.5 Designing Policies

The policy $X^\Pi(S_t)$ will determine the trading action based on the current state of the portfolio. Once a policy has been chosen a Z-score threshold rule will make the decisions. Then when the Z-score on time t exceeds $+Z_{threshold}$ the strategy opens shorting the spread and when it falls below $-Z_{threshold}$ then a long spread position will be opened. Also, when the spread reverted to the mean positions should close. This process will be detailed further.

4.3.6 Evaluating Policies

After implementing the policy, its effectiveness will be evaluated through the backtesting on the training dataset. The evaluation includes metrics such as Sharpe, Sortino and Calmar ratios, maximum drawdown, profit factor and win rate across the period. Nevertheless, the objective in this case will be the Sortino Ratio and the best policy will be the one that maximizes the Sortino Ratio on the training set, since this ratio will consider the expected value of the portfolio over the downside risk of the. All this metrics are detailed further.

4.4 The sequential decision process

This process follows continuous learning cycle, that thanks to the Kalman Filter, the trading strategy can adapt dynamically to new market information. First it requires an initial prediction which is defined as $w_0 = \left[0, \frac{y_0}{x_0}\right]$, then the Kalman Filter projects the next state $(\hat{\alpha}_t, \hat{\beta}_t)$ based on the previous estimates. When the new prices X_t and Y_t at time t it starts the updating step and generating new estimations.

Once the updated state is known, the policy comes in an is time to make new decisions based on the current Z-score. Then the chosen decision is executed, and this will affect the portfolio state. Then the model learns based on all past predictions and starts the recursiveness by incorporating new information to it. This framework ensures continuous adaptation to the market conditions while maintaining stable.

4.5 Q and R matrices selection

In this project, the Q and R matrices were chosen to balance adaptability in the Kalman Filter's dynamic estimation process. In this case, the process noise covariance $Q = 0.0001$ reflects the expected noise on the estimation allowing it to adjust over time. Then the measurement noise covariance $R = 1,000$ represents the uncertainty in the market price observations, this will adjust the trust on the predictions. In addition, there is $P = 0.0001$ which is the initial estimate covariance and ensures that the model begins with high confidence in its initial state estimates.

4.6 Worked example showing state evolution over several periods

For better understanding how the Kalman Filter and the strategy will operate based on the SDA framework proposed, this section presents an illustrative example showing a state evolution over 8 days for a pair. The purpose of this to demonstrate how the system will update and take decisions based on the predictions and information arriving. By tracking the Z-scores it's easier to appreciate the SDA process. The initial state will be $w_0 = [0, 1.5]$, and for easy understanding the thresholds are to open a position when Z-score $> \pm 1.5$ and exit when $|Z\text{-score}| < 1$. And assuming already computed μ_t and σ_t , the example is the following:

Day	X	Y	$\hat{\alpha}_t$	$\hat{\beta}_t$	Spread	μ_t	σ_t	Z-score	Signal	Decision
1	320	610	0	1.5	$610 - (0 + 1.5 \cdot 320) = 130.0$	128.5	8	0.19	0	Flat
2	322	616	0.02	1.48	$616 - (0.02 + 1.48 \cdot 322) = 138.5$	129.2	8.2	1.13	0	Flat
3	325	628	0.04	1.47	$628 - (0.04 + 1.47 \cdot 325) = 149.9$	130.1	8.4	2.36	-1	Short
4	324	625	0.05	1.46	$625 - (0.05 + 1.46 \cdot 324) = 149.1$	131	8.5	2.13	-1	Hold
5	323	620	0.07	1.45	$620 - (0.07 + 1.45 \cdot 323) = 149.3$	131.5	8.6	2.07	-1	Hold
6	326	618	0.09	1.44	$618 - (0.09 + 1.44 \cdot 326) = 148.7$	132	8.7	1.92	-1	Hold
7	330	615	0.1	1.43	$615 - (0.10 + 1.43 \cdot 330) = 142.1$	132.4	8.9	1.09	0	Exit
8	331	612	0.12	1.42	$612 - (0.12 + 1.42 \cdot 331) = 140.5$	132.7	9	0.86	0	Flat

5. Kalman Filter Implementation

A Kalman Filter is a recursive Bayesian estimation algorithm design to estimate a dynamic system in the presence of noise and uncertainty. In the context of this project, this model will provide a robust method to estimate parameters that change over time, in this case, the hedge ratio. Unlike other static this model doesn't assume constant relationships between variables, but rather continuously updates its estimates as new information is known. Also, this model only depends on the last estimation and the new observation, this makes it extremely computationally light and fast to run, since it doesn't depend on a big historic dataset like other machine learning models do.

5.1 Initialization procedures

In this project, the Kalman Filter is initialized to dynamically estimate the hedge ratio between the selected cointegrated pair MSFT and INTU. To initialize the model, an initial estimation of $w_0 = [\alpha_0 \ \beta_0]$ is first required. For this first estimation, $\alpha_0 = 0$ and $\beta_0 = \frac{y_0}{x_0}$. In addition, P_0 which is the initial estimate covariance and ensures that the model begins with high confidence in its initial state estimates is $P_0 = 0.001 \cdot A$, where A is an identity, creating a diagonal matrix with 0.001 on the it. As P_0 increases the Kalman Filter trust less on the first estimation. As P_0 decreases the Kalman Filter trust more on the first estimation. In this case it is given a lot of trust to the first estimation.

Then the observation matrix is defined as $C_t = [1 \ x_t]$, representing the relationship between the observed price and the state vector. The observation matrix is the same to X when a linear regression is being defined on its matrix form, 1 represents the bias term and x_t the price observed for x at time t . With this, is possible to initialize C_0 as $C_0 = [1 \ x_0]$.

5.2 Parameter estimation methodology

For parameter estimation, first it is necessary to calculate the w_t and P_t predictions, as follows:

$$w_{t+1} = Fw_t$$

$$P_{t+1} = FP_tF^T + Q$$

Where:

- F : is an identity matrix
- Q : is the process noise covariance, which is the rate at the filter adapts to new data. While increasing Q , the sensitivity to new movements on the market increases. With Q decreasing, the estimation gets more stable and loose sensitivity.

Then the updating step is performed, and this implies to calculate the Kalman Gain for updating w_t and P_t . This step is performed as follows:

$$K_t = P_{t+1}C^T(CP_{t+1}C^T + R)^{-1}$$

$$w_t = w_{t|t-1} + K_t(y_t - Cw_{t|t-1})$$

$$P_t = (I - KC)P_{t|t-1}$$

In the equations above equations, K_t denotes the Kalman Gain, which will assign the weight to the new observations versus prior estimations. The observation noise covariance R reflects the uncertainty of the market measurements, so by R being increased, implies the observations are noisy and the model will trust more the past predictions over the new data. Decreasing R trusts more on recent observations and trust less on past predictions. For this project parameter values were selected as $Q = 0.001$ and $R = 1,000$. These parameters were found by trial and error by hand for better performance on this pair.

5.3 Reestimation schedule and validation approach

Unlike static regressions that require manual and periodic rebalancing, the Kalman Filter rebalance and reestimates α_t and β_t every time new information comes. Knowing this, each trading day the algorithm processes new prices, computes new predictions and immediately updates the hedge ratio through the recursive step. Also, the rolling estimation for μ_t y σ_t ensures that calculated spreads reflect current market conditions.

5.4 Convergence analysis and filter stability

For a good diagnostic in Kalman Filter models is required an examination of the parameter convergence and stability over time. Illustration 3 shows the evolution of the estimated α_t and β_t throughout the backtesting period for the MSFT-INTU pair. These parameters represent the dynamic relationship between these two assets, where β_t is the hedge ratio and α_t captures any constant drifting on the spread. Overall, the behavior of both parameters validates robustness and the model adapting to new information while maintaining smoothness and not showing a lot of noise.

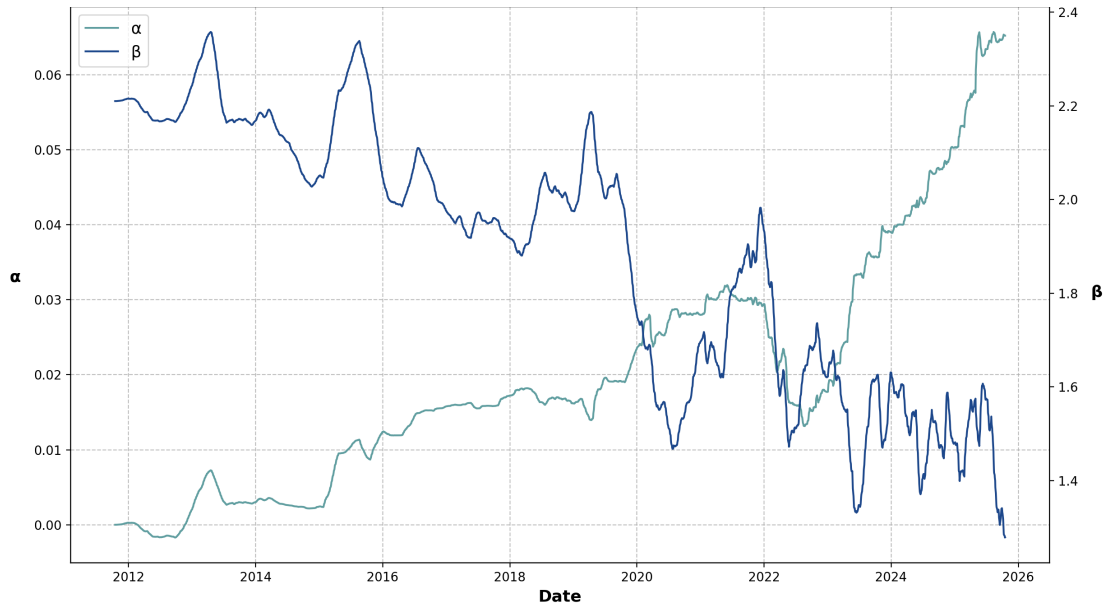


Illustration 3: Plot of the α_t and β_t estimations for the MSFT-INTU pair throughout the backtesting. (Source: own elaboration).

6. Trading Strategy Logic

6.1 Z-Score definition

As said before, the spread is defined as follows:

$$\text{spread}_t = y_t - (\alpha_t + \beta_t x_t)$$

Where x_t and y_t are the prices of MSFT and INTU at time t , and α_t and β_t are the Kalman Filter estimations for the intercept and hedge ratio at time t . The spread will be the residuals of the estimation. For monitoring mean reversions is necessary to standardize the spread, and for this is necessary to compute the Z-score based on the mean and standard deviation of the spread. Since the standard deviation is not specifically expected to be constant over time, this two metrics are going to be calculated with a rolling window 252 days (1 stock market year) and this will help the model to adapt to variations over time. Then the Z-score will be calculated as follows:

$$z_t = \frac{\text{spread}_t - \mu_t}{\sigma_t}$$

The value of this Z-score means how many standard deviations the spread is from its mean. For instance, if the Z-score at time t is 1.5, this will indicate that the spread is 1.5 standard deviations from its mean, and this may indicate trading opportunity since the mean is expected to revert and come back down. Then a short position on the spread is issued.

6.2 Optimal entry and exit Z-score policy found

To find the best Z-score policy, several backtests were performed just on the training set with different Z-score thresholds. To choose the best policy, the optimal threshold was taken as the one that obtained the highest Sortino ratio on this set. With that the selected Z-score policy was a threshold of ± 0.5184 . This implies that when the actual z_t goes over the 0.5184 a short signal for the spread is issued. However, if the actual z_t goes below -0.5184 then a long signal for the spread is issued. All this expecting that the actual z_t will revert to its mean.

6.3 Cost treatment: commissions and borrow rates

The backtest performed applies real commission and borrow costs defined as follows:

- **Commissions:** a rate of 0.125% is applied to the entry and exit of each trade. It is charged on each stock and doesn't matter if it's a long or a short and is immediately discounted from the actual cash.
- **Borrow costs for short trades:** For every day the position remains open, an annual rate of 0.25% (or 0.25%/252 daily) of the current market value of the position is discounted from the actual capital.

7. Results and Performance Analysis

7.1 Performance metrics

For evaluating the strategy results and performance, several metrics will be calculated to consider both return and risk. These metrics are key to identify on which periods the strategy perform best, as they can be compared one to another to highlight strengths and weaknesses in each period. For the metrics that require annualization, an annualizing factor of 252 will be used, since the project is based on daily data and the market is opened 252 days of the year.

7.1.1 Sharpe Ratio

The Sharpe Ratio is one of the most well-known and used risk-adjusted performance metrics, as it indicates how much return the portfolio generates for each unit of risk. This ratio is calculated as follows:

$$\text{Sharpe} = \frac{\mu_{rets}}{\sigma_{rets}}$$

- μ_{rets} : Is the annualized mean of the returns.

- σ_{rets} : Is the annualized standard deviation of the returns.

7.1.2 Sortino Ratio

In general terms, this metric is very similar to Sharpe Ratio, however, the Sortino Ratio does not consider the total volatility of the portfolio. Instead, it only considers the negative volatility (downside risk), meaning that only losses are penalized. This ratio is calculated as follows:

$$\text{Sortino} = \frac{\mu_{rets}}{\sigma_{drets}}$$

- μ_{rets} : Is the annualized mean of the returns.
- σ_{drets} : Is the annualized downside volatility of the returns.

7.1.3 Maximum drawdown

This metric measures the largest percentage drop from a local maximum (peak) of the portfolio to the local minimum (trough). It evaluates how severe are the potential losses an investor might experience. It is calculated as follows:}

$$\text{MDD} = \frac{\text{Peak} - \text{Trough}}{\text{Peak}}$$

7.1.4 Calmar Ratio

This metric relates the portfolio's return to its maximum drawdown. It provides a risk adjusted measure of performance focused on the risk of large losses. A higher Calmar Ratio indicates that the strategy achieves high returns with small drawdowns. It is calculated as follows:

$$\text{Calmar} = \frac{\mu_{rets}}{\text{MDD}}$$

- μ_{rets} : Is the annualized mean of the returns.
- MDD: Is the maximum drawdown.

7.1.5 Profit Factor

The profit factor is a performance metric that measures the ratio of total profits over the total losses generated by a trading strategy. This provides an evaluation of its profitability, and it's calculated as the sum of all winning trade profits divided by the absolute sum of all losing trade losses. A profit factor that is greater than 1 indicates

that the strategy generates more profit than losses, therefore a factor below 1 is an unprofitable strategy. Usually, a strategy with profit factor over 1.75 is considered as a strong performance strategy. It is calculated as follows:

$$\text{Profit Factor} = \frac{\sum \text{Profits from winning trades}}{|\sum \text{Losses from all losing trades}|}$$

7.2 Trade statistics

7.2.1 Win ratio

This is a simple measure representing the proportion of trades that generated a profit relative to the total number of trades. It tells how likely the strategy was to achieve winning trades. It is calculated as follows:

$$\text{Win Ratio} = \frac{\text{Number of winning trades}}{\text{Total Trades.}}$$

7.2.2 Average return per trade

This statistic represents the mean of all returns separated by long and short operations. This indicates how profitable in percentage terms is the average trade after commission and borrow costs. It is calculated as follows:

$$\text{Average return per trade}_{\text{LONG}} = \frac{1}{n} \sum_{i=1}^n \frac{\text{Exit price}_i - \text{Entry price}_i}{\text{Entry Price}_i}$$

$$\text{Average return per trade}_{\text{SHORT}} = \frac{1}{n} \sum_{i=1}^n \frac{\text{Entry price}_i - \text{Exit price}_i}{\text{Entry Price}_i}$$

7.3 Equity curve plots (train, test, validation periods)

Illustration 4 shows the evolution of the portfolio value starting with 1 million USD throughout the training, testing, and validation periods. It offers a view of the strategy's cumulative performance under real trading conditions. During the training phase, the portfolio shows a consistent upward trend, suggesting strong profitability and effective signal generation under the historical dynamics used for calibration.

The test phase, in contrast, shows a bad performance for the strategy, nevertheless it's important to note that's when the Covid19 pandemic occurred and may cause changes in the MSFT–INTU relationship. However, during the validation period, the portfolio showed a robust growth trend, steadily increasing to reach approximately 3.6

million USD at the end of the backtest. This performance underscores the model's capacity to adapt and remain profitable in out-of-sample conditions.

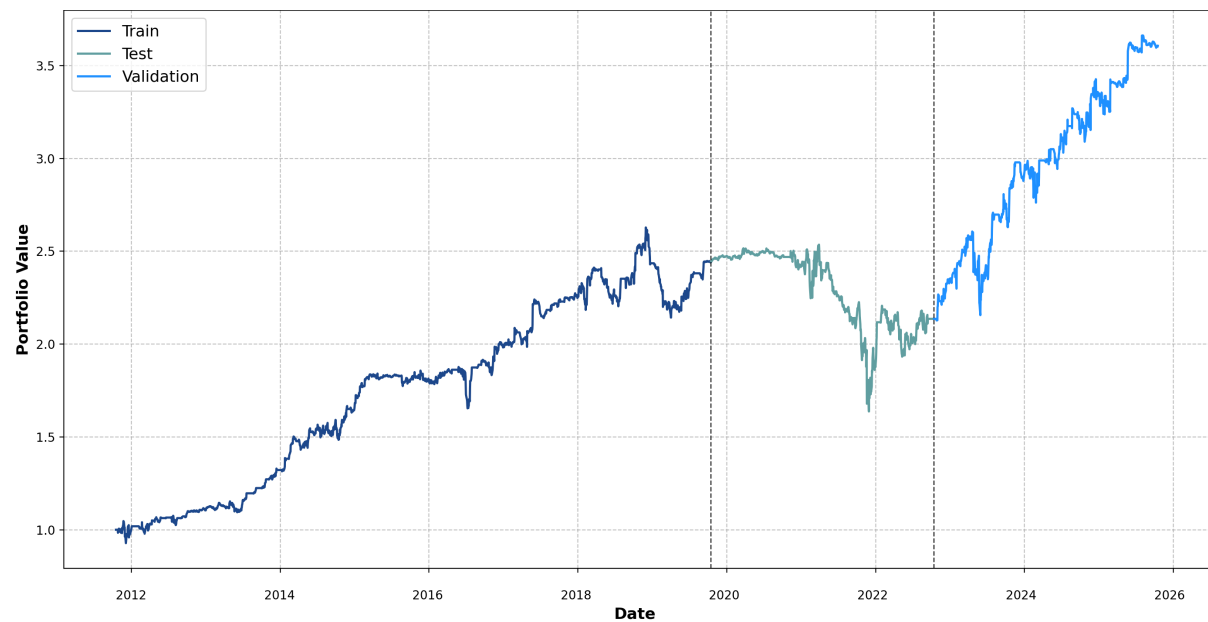


Illustration 4: Total equity evolution over train/test/validation periods. (Source: own elaboration).

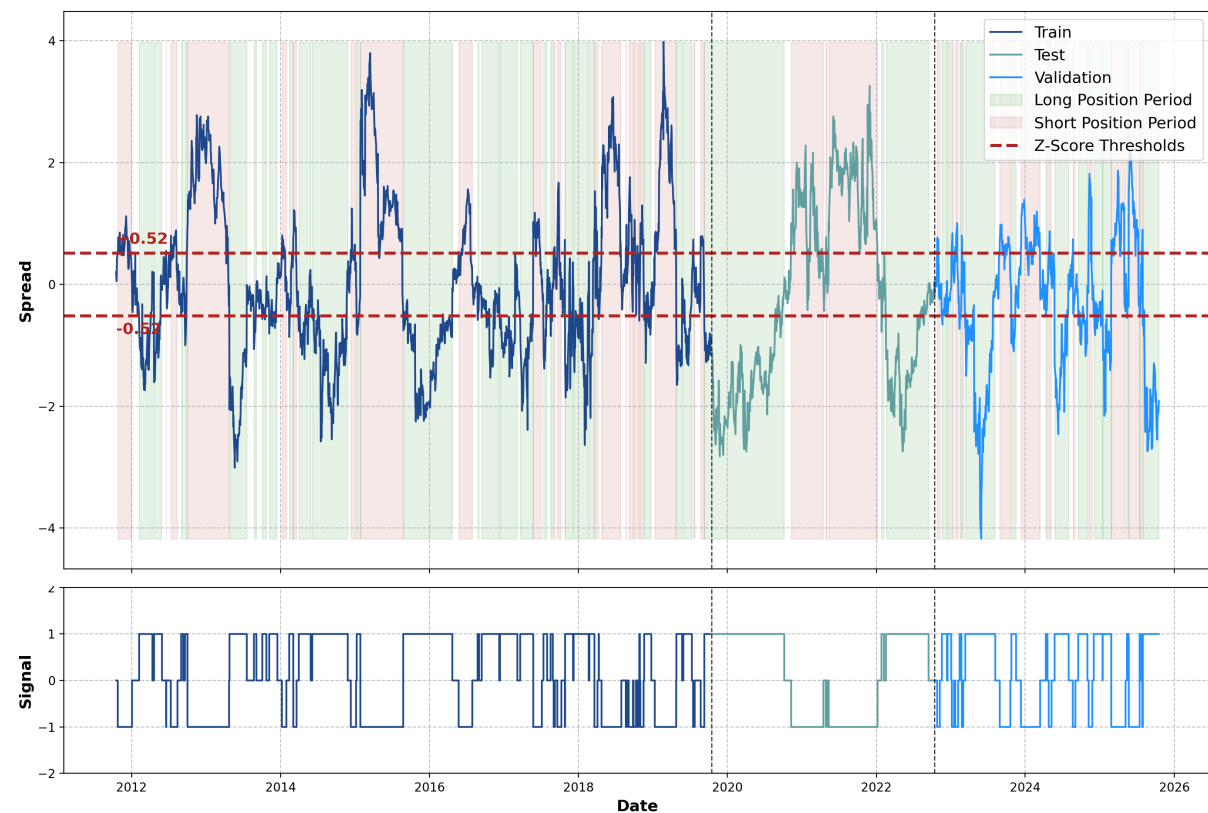


Illustration 5: Standardized spread and trading signals over train/test/validation periods. (Source: own elaboration).

Illustration 5 shows the evolution of the standardized spread (z_t) over training, testing and validation periods and below are the generated signals over the backtest. Also,

on the spread graph the policy Z-score threshold at ± 0.52 . This illustration helps to visualize all the SDA and Kalman Filter implementation for the pairs trading strategy: whenever the estimated spread deviates significantly from its mean decisions are taken waiting for mean reversion.

7.4 Performance on train, test and validation periods and cost analysis.

7.4.1 Performance on train period

As shown in the table below, during the training period, the strategy between the MSFT and INTU pair had a solid performance indicating that the model effectively captured the mean-reversion patterns within the spread. This achieved a Sharpe and Sortino ratios of 1.06 and 1.32 respectively a good risk-adjusted performance. In addition, it obtained a Calmar ratio of 0.63 which is decent and showed that the strategy maintained stability. Also, in all the approximately 8 years of training just had a maximum drawdown of 18.47% and means that is not expected big drawdowns, because at the end the strategy had a strong win rate and a profit factor of 1.87 meaning that portfolio almost doubles the losses.

Metric	Value
Sharpe Ratio	1.0654
Sortino Ratio	1.3196
Maximum Drawdown	18.47%
Calmar Ratio	0.6388
Number of Long Trades	106
Number of Short Trades	106
Total Borrowed Amount	\$15,371.65
Total Commission Cost	\$236,166.56
Win Rate (Long Positions)	82.08%
Win Rate (Short Positions)	48.11%
Overall Win Rate	65.09%
Average Return per Long Trade	4.74%
Average Return per Short Trade	-1.66%
Average Return per Trade	1.54%
Profit Factor	1.868

7.4.2 Performance on test period

During the testing period the strategy experience a bad moment compared to the training period and revealed that didn't adapt that good to the market conditions. It got negatives Sharpe, Sortino and Calmar ratios, and also, almost doubled the previous maximum drawdown, suggesting that in this period the strategy was exposed to larger

to more losses. The winrate on this period decreased and the average return per short grew. Finally the profit factor got lower than 1 and means that the losses were bigger than the profits.

Metric	Value
Sharpe Ratio	-0.1313
Sortino Ratio	-0.1428
Maximum Drawdown	35.44%
Calmar Ratio	-0.0725
Number of Long Trades	17
Number of Short Trades	17
Total Borrowed Amount	\$11,904.45
Total Commission Cost	\$52,537.12
Win Rate (Long Positions)	70.59%
Win Rate (Short Positions)	35.29%
Overall Win Rate	52.94%
Average Return per Long Trade	9.18%
Average Return per Short Trade	-6.68%
Average Return per Trade	1.25%
Profit Factor	0.887

7.4.3 Performance on validation period

In the validation period, the strategy got an outstanding performance with a big recovery. This period got the highest Sharpe, Sortino and Calmar ratios demonstrating that the strategy achieved a great risk-adjusted while managing downside risk. Also, the maximum drawdown was lower than the one of the train periods indicating better stability. The overall win rate also got higher than train period and the profit factor of 2.23 suggests that total profits more than doubled the total losses reinforcing efficiency.

Metric	Value
Sharpe Ratio	1.1816
Sortino Ratio	1.5154
Maximum Drawdown	17.29%
Calmar Ratio	1.0844
Number of Long Trades	51
Number of Short Trades	51
Total Borrowed Amount	\$9,919.83
Total Commission Cost	\$183,400.95
Win Rate (Long Positions)	70.59%

Win Rate (Short Positions)	60.78%
Overall Win Rate	65.69%
Average Return per Long Trade	4.70%
Average Return per Short Trade	-0.81%
Average Return per Trade	1.95%
Profit Factor	2.225

7.4.4 Performance overall sets

Performing the backtest over the full period the pairs trading strategy demonstrated consistent profitability over time with moderate volatility with occasional important drawdowns, the Sharpe, Sortino and Calmar ratios are decent considering that this was a long time backtest. Achieving a winrate of 64% also shows good performance and specifically on the long trades. The profit factor of 1.63 also is a great indicator that total profits outweighed the losses by more than 60%. All of this validates the robustness of the model and the capacity of adapting of the Kalman Filter over long time periods with its dynamic hedging approach.

Metric	Value
Sharpe Ratio	0.7119
Sortino Ratio	0.8205
Maximum Drawdown	37.71%
Calmar Ratio	0.2709
Number of Long Trades	178
Number of Short Trades	178
Total Borrowed Amount	\$38,773.46
Total Commission Cost	\$480,060.61
Win Rate (Long Positions)	76.40%
Win Rate (Short Positions)	51.69%
Overall Win Rate	64.04%
Average Return per Long Trade	5.16%
Average Return per Short Trade	-1.90%
Average Return per Trade	1.63%
Profit Factor	1.642

7.4.5 Cost analysis and portfolio summary

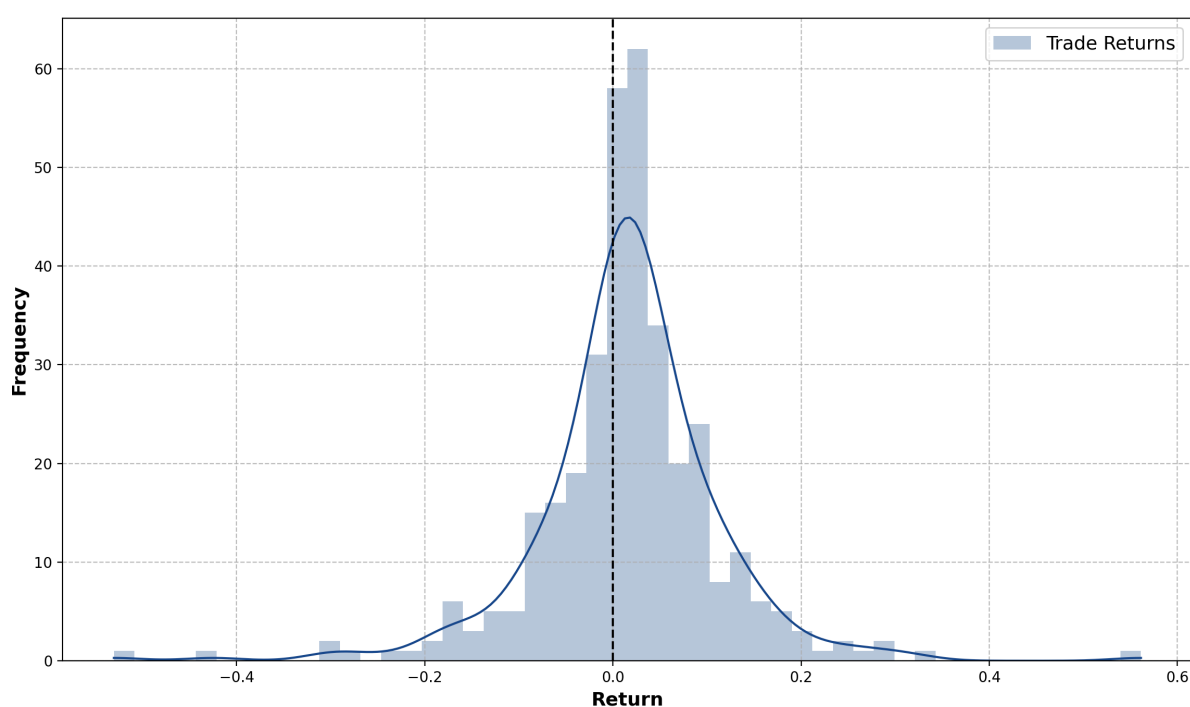
While analyzing costs, the strategy maintained profitability despite the significant expenses confirming economic feasibility. Throughout the entire backtest period total commission costs were of \$480,060.61 and total borrow costs were of \$38,773.46 while profit was \$2,606,857.73, so there's no preoccupation about the costs at all.

Finally the total return on investment (ROI) of 260.69% demonstrated the positive results of the strategy.

Metric	Value
Initial Capital	\$1,000,000.00
Final Capital (Train)	\$2,440,923.55
Final Capital (Test)	\$2,136,025.78
Final Capital (Validation)	\$3,606,857.73
Total ROI	260.69%
Total Borrow Cost Paid	\$38,773.46
Total Commission Paid	\$480,060.61

7.4.6 Distribution of returns per trade

After performing the overall periods backtest, the individual return per trade was saved for trade statistics, and, for plotting reasons. The graph bellow shows frequency histogram of individual trade returns. This histogram shows that the strategy got more trades with positive returns than negative since the distribution is skewed, and this is good for the strategy, meaning that there's more probability to win trades than losing them. Finally some important outliers are shown, with one single return of almost 60% and a loss of approximately 50%.



8. Conclusions

This project demonstrates that a pairs trading strategy with a deep cointegration analysis and a Sequential Decision Analysis (SDA) with Kalman Filters approach for dynamic estimations can generate persistent and solid profits. This strategy performed a statistical arbitrage controlling the risks and uncertainty while producing robust results, especially in the strong recovery over the validation set. The pair of MSFT and INTU pair delivered a final validation capital of \$3.61 M USD from a \$1M USD start despite all the commission and borrowing costs, making it attractive and feasible.

In addition, risk-adjusted metrics show great viability over this strategy, having a Sharpe, Sortino and Calmar ratios of 1.18, 1.52 and 1.08 in the validation set with a profit factor of 2.23 and overall 66% operation winrate. These metrics are strong evidence that the strategy had good dynamic adjustments and estimations, making possible to generate real winning signals waiting for mean reversion processes. Also, in terms of costs, yes, they were high, and yet still achieved a profit factor of 1.64 and a total ROI of 260% over all the periods.

Over the test period the strategy had a bad performance, but it's important to remember that this was a bad period for all stocks because of the Covid19 pandemic. While this strategy should make profits on downward trends, the pandemic could have changed the relationship between these assets for a short period of time, since there was a recovery on the validation set.

There are some potential improvements for this project, like implementing a market regime check or dynamic cointegration tests, making the policy a lot more complex and adding new decisions like stopping the strategy while waiting for new cointegration signals. Also, a dynamic reestimation for the Z-score threshold may help to take better decisions while adapting to new uncertainty.

Finally, it may be interesting to test the performance of a portfolio with several cointegrated pairs while optimizing capital allocation for maximizing specific ratios like Sortino or Calmar and dynamically estimate hedge ratios for each pair. This strategy may diversify a lot more the risks and limiting drawdowns.

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